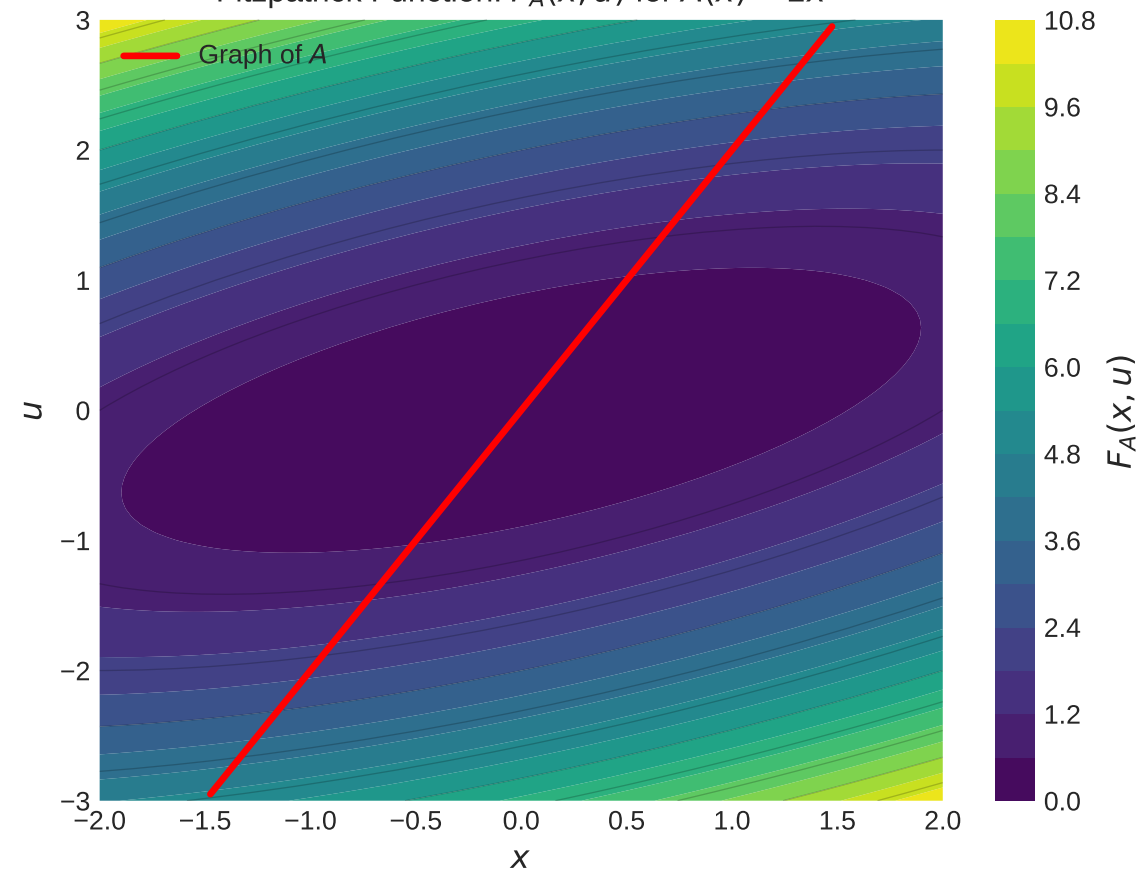


Fitzpatrick Function: $F_A(x, u)$ for $A(x) = 2x$



Key Properties of F_A :

(i) $F_A(x, u) = \langle x \mid u \rangle - \inf_{(y, v) \in \text{gra}A} \langle x - y \mid u - v \rangle$

(ii) $F_A(x, u) \geq \langle x \mid u \rangle$ for all (x, u)

(iii) $(x, u) \in \text{gra}A \Rightarrow F_A(x, u) = \langle x \mid u \rangle$

(iv) $F_A \in \Gamma_0(\mathcal{H} \oplus \mathcal{H})$ when A is m.m.

(v) $\text{gra}A = \{ (x, u) \mid F_A(x, u) = \langle x \mid u \rangle \}$